

Samuel Zhang

Applied AI | Machine Learning | Product Engineering

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Profile

Applied AI builder and product owner completing an MSc in AI Applications and Innovation at Imperial College London. Combines machine-learning research, experimental design and production engineering with the stakeholder work needed to turn ambiguous problems into trusted products. Built enterprise software at Pfizer with measurable operational impact, develops evidence-bounded AI for insurance, and founded COVERD to create adaptive, human-owned recruitment interviews.

Core Expertise

AI/ML: PyTorch, TensorFlow, scikit-learn, LLM systems, ranking and evaluation, document processing, computer vision, statistical modelling, experiment design

Engineering: Python, Julia, SQL, TypeScript, JavaScript, MATLAB, React/Next.js, APIs, PostgreSQL, Docker, Linux, CI/CD, GPU and HPC systems

Product: Discovery, requirements, roadmaps, rapid prototyping, A/B testing, stakeholder management, responsible AI, technical documentation, adoption and training

Selected Experience

Marsh Risk | *AI Intern – Data & Machine Learning*

May 2026 – Present

London, United Kingdom

- Investigating how historical placement data and contract wording can support better-informed decisions in the London insurance market without treating historical patterns as present-day truth.
- Building and stress-testing ranking, document-processing and contract-review prototypes with temporal evaluation, explicit provenance and visible uncertainty.
- Translating research findings into bounded decision support: separating evidence sources, documenting limitations and identifying when a system should abstain rather than overstate confidence.

coverd.ai | *Founder*

Mar 2026 – Present

London, United Kingdom

- Building “the AI interviewer that interviews the company first”: team interviews capture role knowledge before candidates are assessed, including success criteria, common pitfalls and capability gaps.
- Designed a role-specific belief graph that updates as evidence arrives, selects the next useful voice-interview question and returns recruiter-reviewable evaluations cited to candidate answers.
- Tested three interview-agent architectures with users and used comparative evidence to choose the product direction; developing the bootstrapped prototype with four design partners, including Imperial College London.
- Keeping consequential hiring decisions human-owned by surfacing missing evidence and uncertainty instead of reducing candidates to an opaque score.

Pfizer, Analytical R&D | *Web Application Developer & Product Owner*

Oct 2024 – Apr 2026

London, United Kingdom

- Owned GROWMAT, an enterprise resource, forecasting and analytics platform, from operational discovery and senior-director pitch through roadmap, agile delivery, deployment and department-wide adoption.
- Connected technical capability with real staff availability and selected maintainable tools over fashionable technology, building stakeholder trust through a polished, dependable product.
- Delivered a reported 1,200% efficiency improvement and saved 120+ person-hours per month across 40+ analysts while maintaining 99.9% uptime and improving application performance by 80%.
- Built continuous analytics for timelines, compliance, analytical performance and future workload; produced API documentation, user guides and video training to support long-term ownership.

Pfizer, Analytical R&D | *Data Analyst Undergraduate*

Sep 2023 – Aug 2024

Sandwich, United Kingdom

- Prototyped pharmaceutical solubility modelling using statistical thermodynamics, Julia and machine-learning libraries to reduce experimental testing requirements.
- Worked across Pfizer, MIT and Imperial College London to translate open-source fluid-thermodynamics research into a practical drug-development workflow.

Research, Teaching & Public Service

King's College London | *Coding Series Tutor & Curriculum Designer* **Jan 2023 – Apr 2025**

London, United Kingdom

- Designed and delivered 20+ programming, data-analysis and introductory ML sessions for 80+ chemistry students, achieving 90% satisfaction and Royal Society of Chemistry endorsement.
- Organised a cross-industry data-science careers panel for 100+ students; mentored learners in using technical skills to widen their career options.

Royal Society / King's College London | *Summer Research Fellow* **Jun 2023 – Jul 2023**

London, United Kingdom

- Built GPU-accelerated GROMACS workflows for protein-membrane molecular simulations, improving computational performance by 70% and iterating experimental design from simulation results.

King's College London | *Undergraduate Research Fellow* **Jun 2022 – Jul 2022**

London, United Kingdom

- Engineered MATLAB, Python and Excel pipelines for rotational spectroscopy, reducing analysis time by 40% and creating a reusable workflow adopted by the research group.

Singapore Civil Defence Force | *Commander's Personal Assistant / Sergeant* **Jul 2019 – Jul 2021**

Singapore

- Built MATLAB-based COVID-19 resource-planning models from public epidemiological data and automated emergency activation attendance across legacy systems for 1,200+ personnel.
- Supported senior leadership in time-critical operations, balancing incomplete information, rapid prioritisation and accountability; received the Service Excellence Award and 1st Division HQ Wall of Fame recognition.

Education

Imperial College London | *MSc AI Applications and Innovation* **Sep 2025 – Sep 2026**

London, United Kingdom

Deep Learning, AI Safety, Innovation Management, Machine Learning in Medical Imaging, Machine Learning in Climate Change

King's College London | *BSc Chemistry with Biomedicine – First-Class Honours* **Sep 2021 – May 2025**

London, United Kingdom

Professional placement; Computational Chemistry, Molecular Biology, Chemical Biology, Organic Chemistry; Associate of King's College London

Selected Technical Projects

- **Home Lab & Private AI Infrastructure** – Designed and operate seven physical servers and 23 services across Proxmox and Docker, including 42 GB VRAM GPU compute, self-hosted GitHub Actions CI, cross-architecture workloads, monitoring, network security and a 96 TB raw / 72 TB usable RAID array.
- **Stock Market Simulation Engine** – Built Julia/SQL order matching, real-time WebSocket order-book updates and a responsive trading interface.
- **COVID-19 Decision Support** – Created predictive analytics and workflow automation for emergency planning and large-scale personnel operations.
- **Molecular Recognition** – Combined microwave spectroscopy with computational predictions through automated MATLAB and Excel analysis workflows.

Awards & Additional Information

Awards: King's Research Experience Award; SCDF Service Excellence Award; SCDF 1st Division HQ Wall of Fame; EARCOS Global Citizenship Award

Languages: English and Mandarin (native/bilingual); Italian (elementary)

Interests: Competitive brass band (euphonium, UniBrass); musical theatre; photography; hiking and climbing